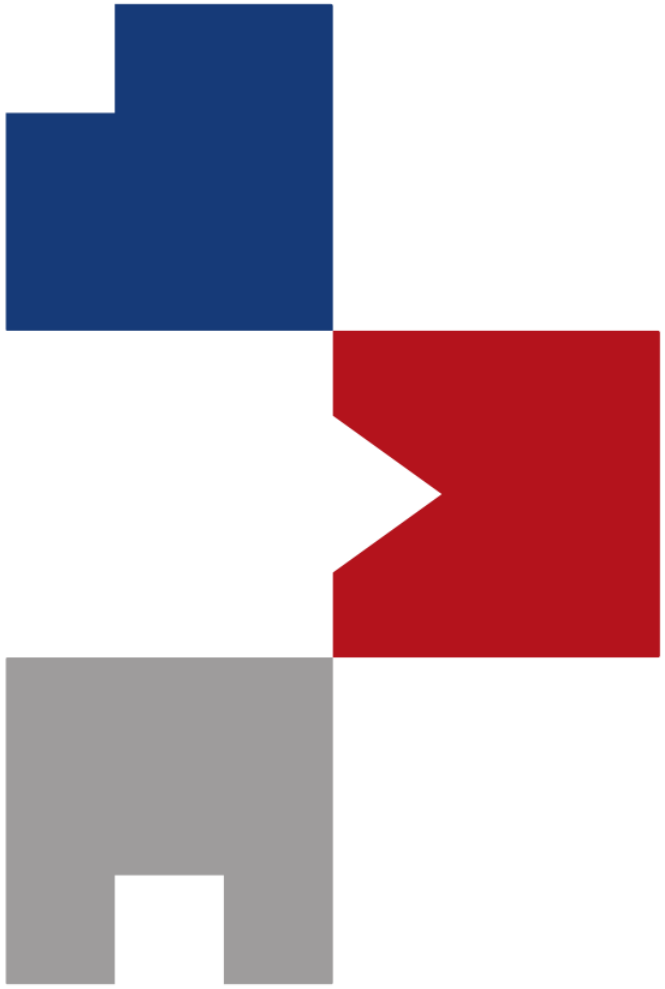


Memory and Storage Devices

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Table of Contents

- **How to design and implement a logic to process information?**
  - Logic gates
  - Combinatorial logic circuits
  - Sequential logic circuits

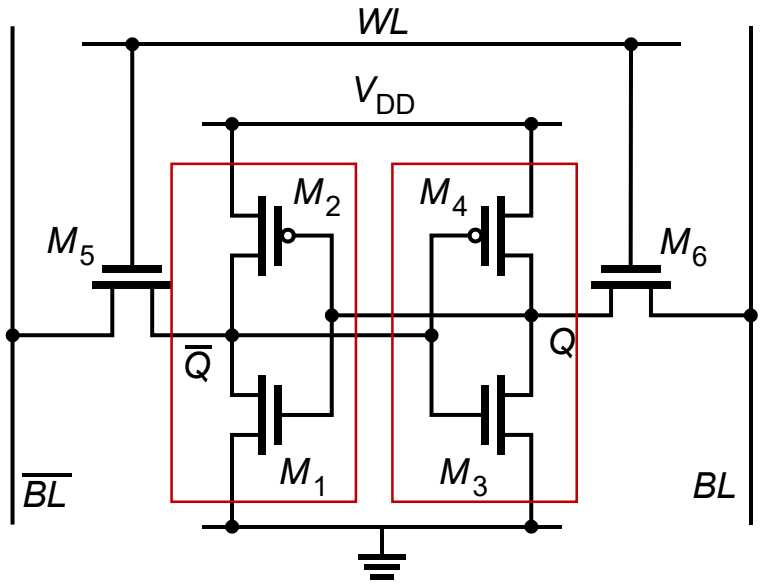
- **How to store the information?**
  - 1-bit: Latches and flip-flops
  - n-bit: Registers
  - Multi-byte: **RAM**, **ROM**, **flash memory**, **HDD**, **SSD**

RAM

- **Random Access Memory (RAM)**
  - A types of memory that can be read and written in any order (~ random access)
  - Typically volatile memory (Note) *volatile* in Korean: 휘발성)
    - The contents of RAM are removed when the power is off.
  - Typically used to store *working* data and ~~programs~~ machine codes
- **Static RAM (SRAM)**
  - Implemented using flip-flops or transistors
    - **Faster read/write speed than DRAM**
  - Typically used for CPU cache, CPU register files, HDD buffers, and router buffers
- **Dynamic RAM (DRAM)**
  - Implemented using capacitors
    - DRAM requires an external memory refresh (periodic reading and rewriting (~ charging) the contents of the DRAM; that is why it is *dynamic*).
    - **Smaller (and denser) and cheaper than SRAM**
  - Typically used for main memory for computers

Static RAM

- **CMOS SRAM**
  - Most common SRAM implementation with low power consumption
  - **A memory cell**: 6 transistors (two cross-coupled NOT gates) [more explanation]
    - Hold state (WL=0)
    - Reading (WL=1): Reading the state at BL
    - Writing: Set (WL=1, BL=1, $\overline{BL}$=0), Reset (WL=1, BL=0, $\overline{BL}$=1)



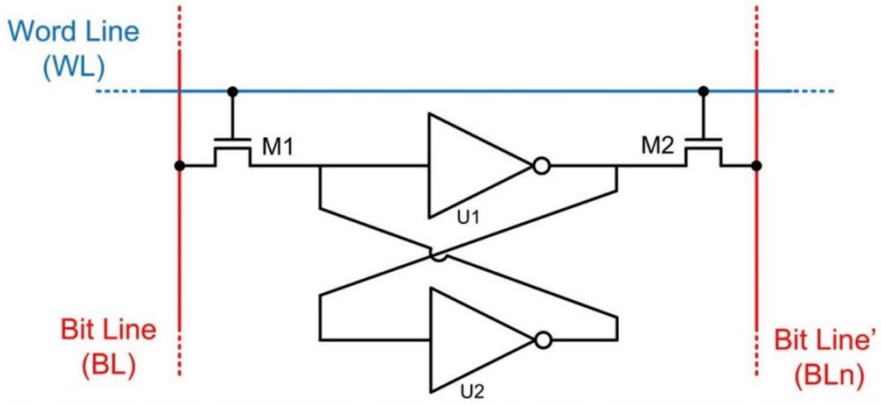


Image: Wikipedia / Brock J. LaMeres, Introduction to Logic Circuits & Logic Design with VHDL, Springer, 2019

Static RAM

- **CMOS SRAM**
  - 4 x 4-bit SRAM array

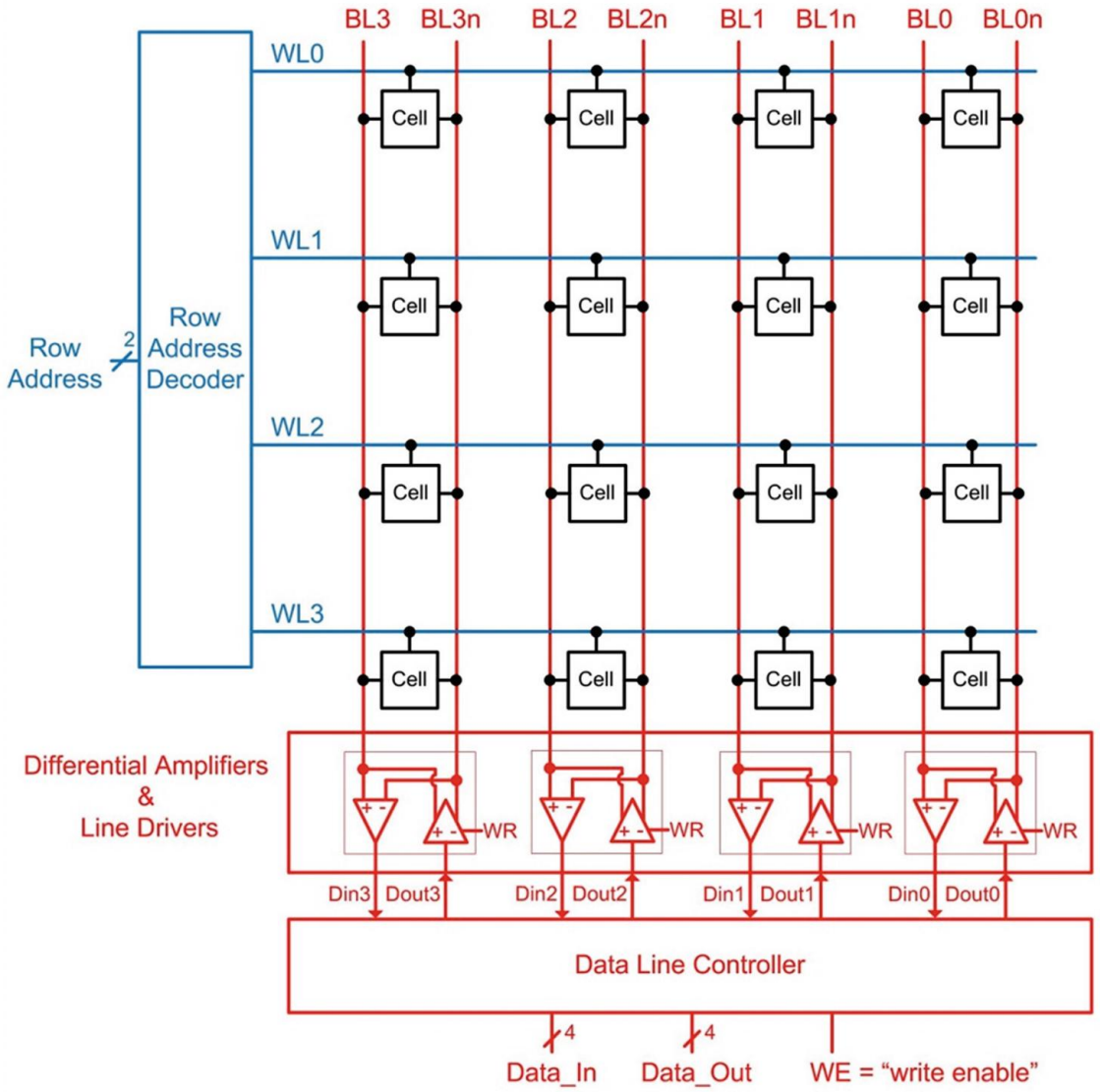


Image: Brock J. LaMeres, Introduction to Logic Circuits & Logic Design with VHDL, Springer, 2019

Static RAM

- **CMOS SRAM**
  - 4 x 4-bit SRAM array

Writing “0111” @ address “01”

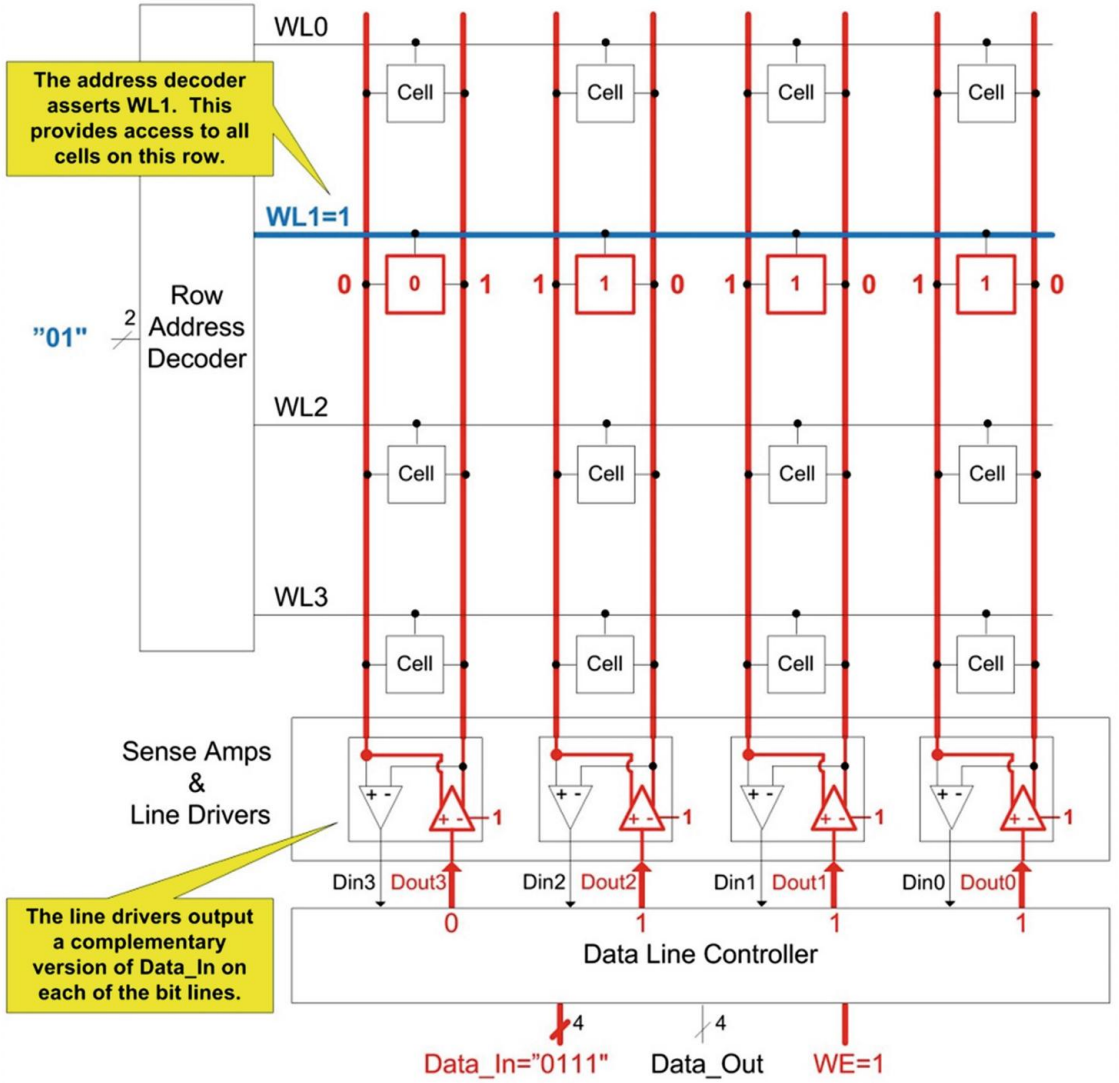


Image: Brock J. LaMeres, Introduction to Logic Circuits & Logic Design with VHDL, Springer, 2019

Static RAM

- **CMOS SRAM**
  - 4 x 4-bit SRAM array

Reading "1010" @ address "10"

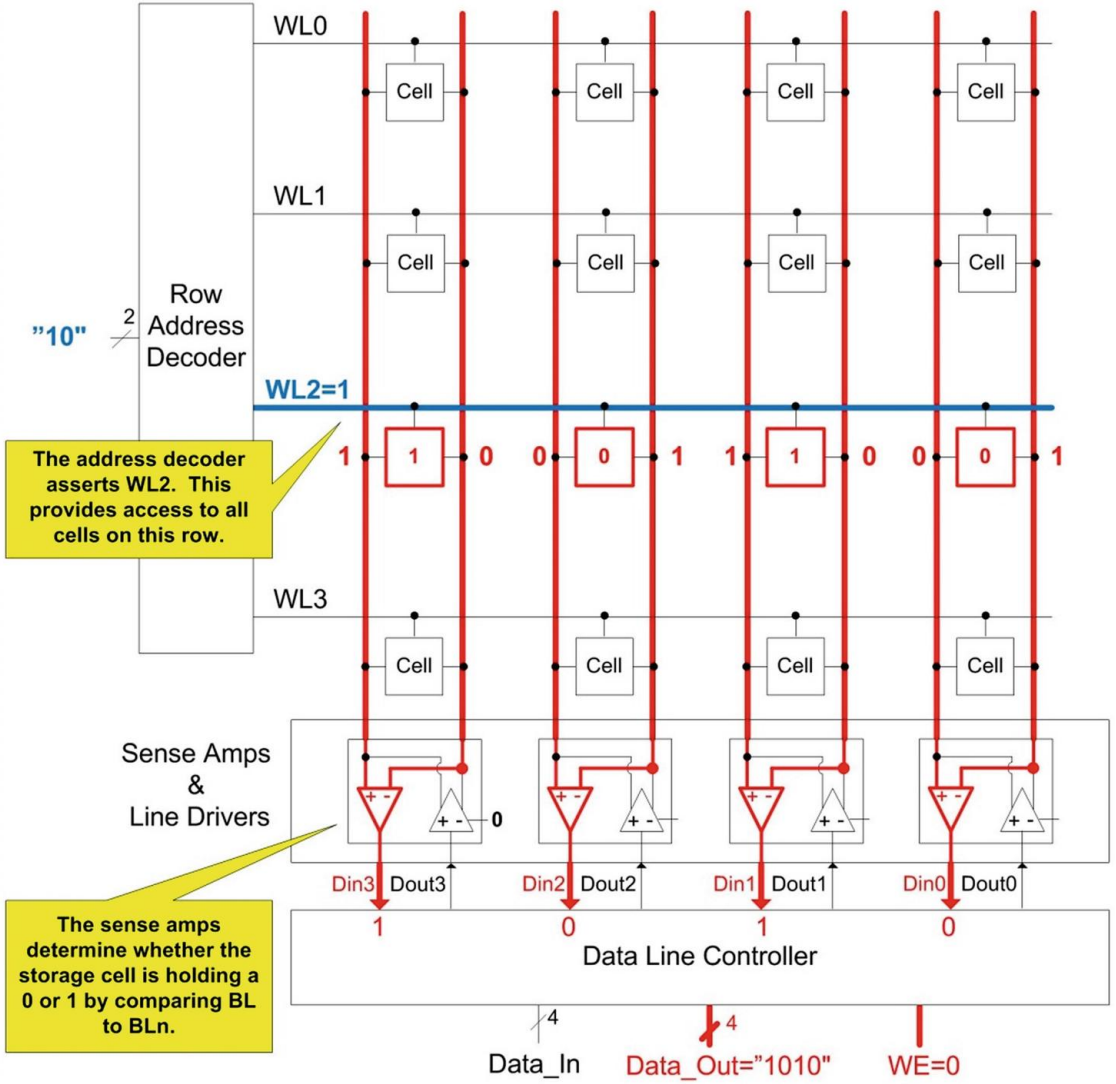


Image: Brock J. LaMeres, Introduction to Logic Circuits & Logic Design with VHDL, Springer, 2019

Dynamic RAM

- **DRAM using capacitors**
  - A memory cell: A capacitor and a transistor [more explanation]
    - Hold state (WL=0)
    - Reading (WL=1): Reading the state at DL
    - Writing
      - Set (WL=1, DL=1; **charging**)
      - Reset (WL=1, DL=0; **discharging**)

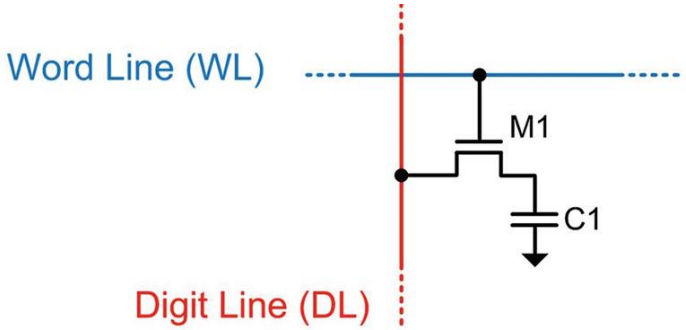


Image: Brock J. LaMeres, Introduction to Logic Circuits & Logic Design with VHDL, Springer, 2019

Dynamic RAM

- **DRAM using capacitors**
  - 4 x 4-bit DRAM array

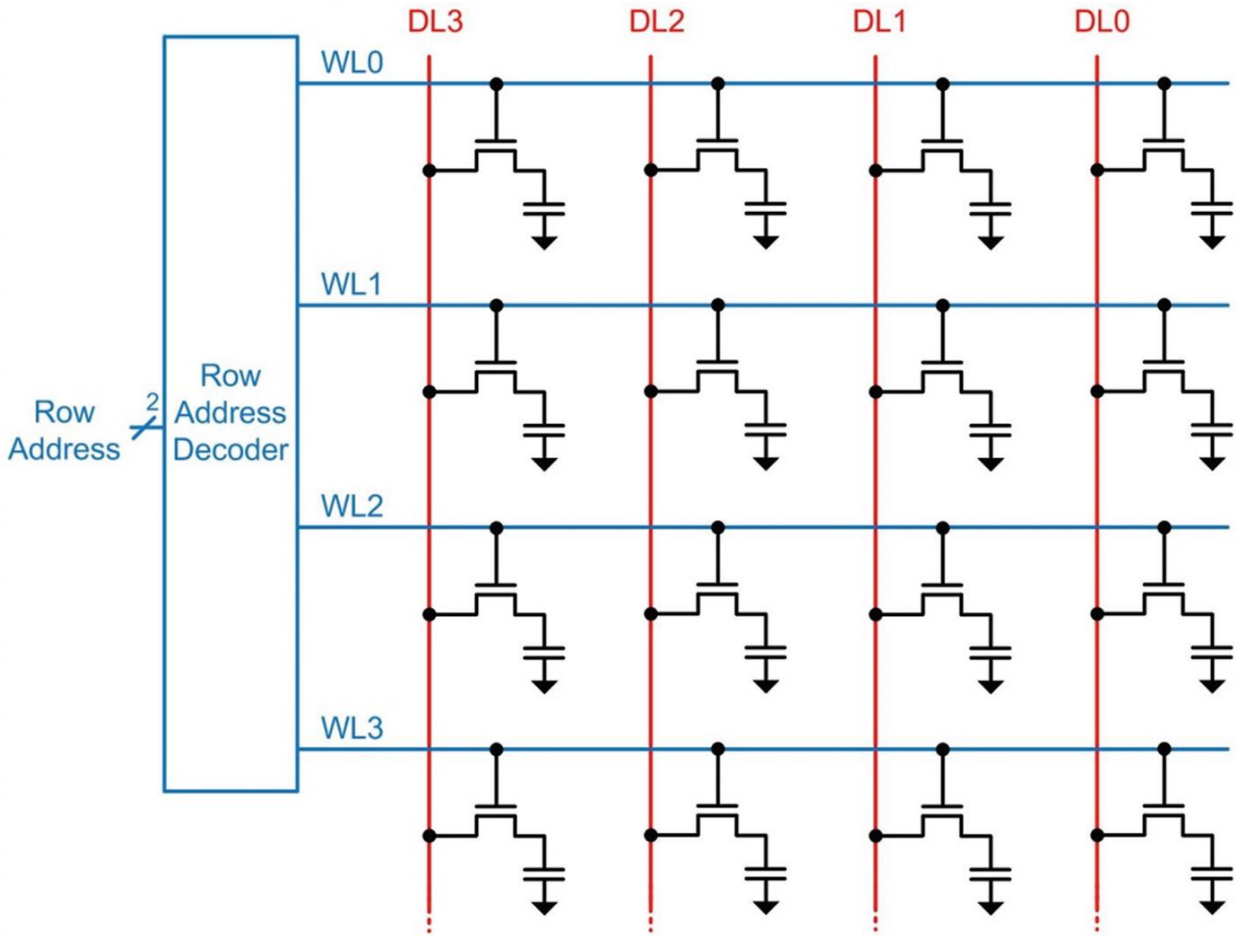
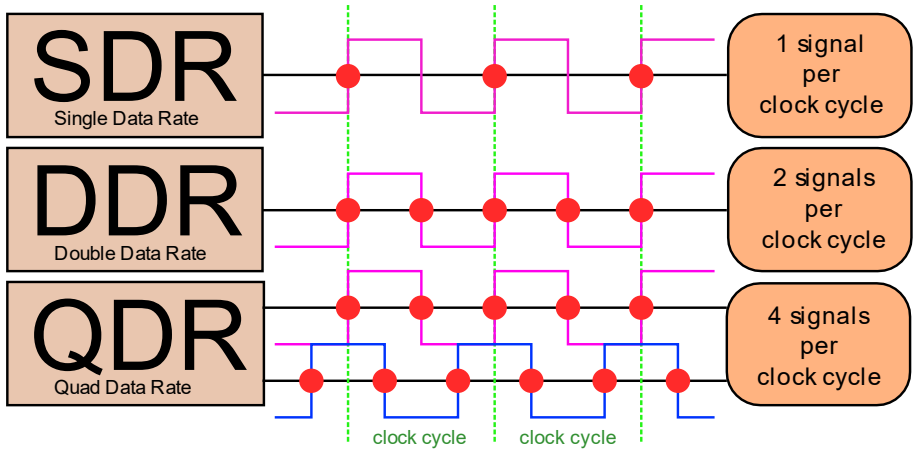


Image: Brock J. LaMeres, Introduction to Logic Circuits & Logic Design with VHDL, Springer, 2019

Synchronous DRAM

- **Synchronous DRAM (SDRAM)**
  - DRAM operates with external *clock signal*
- **SDR SDRAM** (Single Data Rate SDRAM)
  - e.g. PC**133** (year 1997, 3.3V, access 64 bits)
    - Chip (clock: **133** MHz) / Bus (clock: **133** MHz, bandwidth: 1066 MB/s)
- **DDR SDRAM** (Double Data Rate SDRAM)
  - e.g. DDR-**400** / PC-3200 (year 2000, 3.3V, access 64 bits)
    - Chip (clock: **200** MHz) / Bus (clock: **200** MHz, bandwidth: 3200 MB/s)



- Note) **GDDR SDRAM** with additional functionalities for GPUs

Image: Wikipedia

Synchronous DRAM

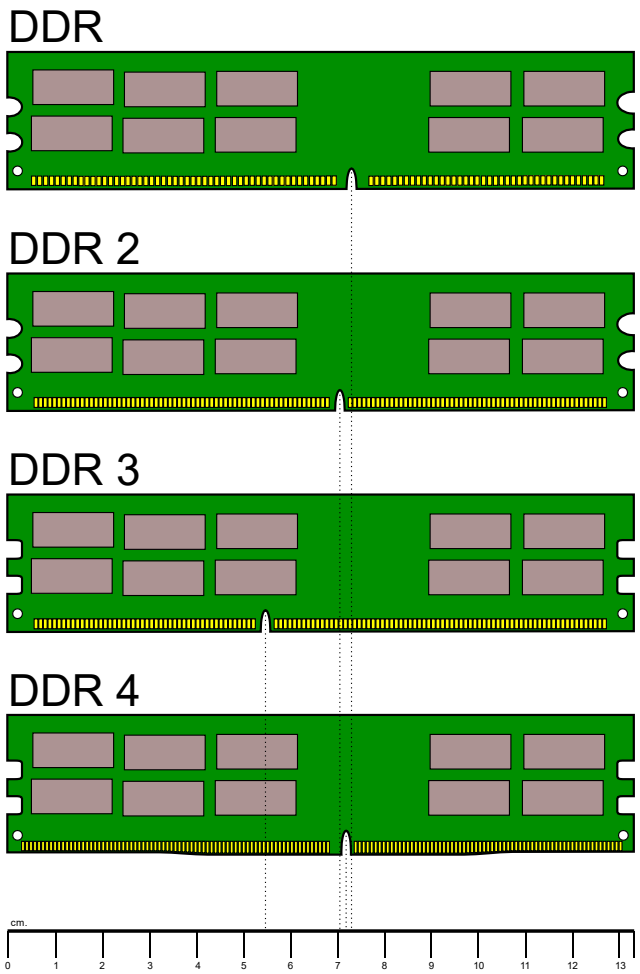
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  - e.g. DDR-**400** / PC-3200 (year 2000, 3.3V, access 64 bits)
    - Chip (clock: **200** MHz) / Bus (clock: **200** MHz, bandwidth: 3200 MB/s)
  - e.g. DDR**2**-**800** / PC**2**-6400 (year 2003, 2.5V, access 128 bits)
    - Chip (clock: **200** MHz) / Bus (clock: **400** MHz, bandwidth: 6400 MB/s)
  - e.g. DDR**3**-**1600** / PC**3**-12800 (year 2007, 1.5V, access 256 bits)
    - Chip (clock: **200** MHz) / Bus (clock: **800** MHz, bandwidth: 12800 MB/s)
  - e.g. DDR**4**-**1600** / PC**4**-12800 (year 2014, 1.2V, access 256 bits)
    - Chip (clock: **200** MHz) / Bus (clock: **800** MHz, bandwidth: 12800 MB/s)

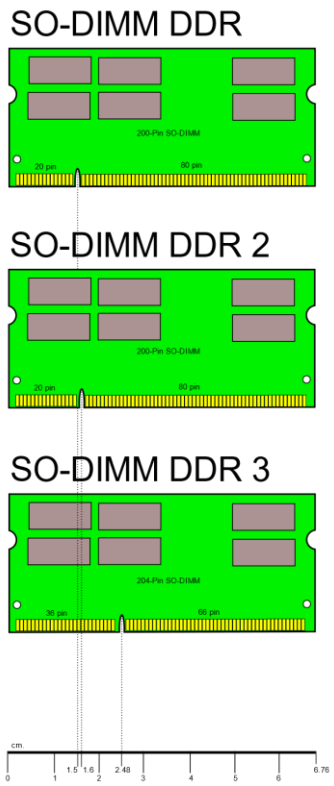
Synchronous DRAM

- **DDR SDRAM form factors**

DIMM (dual in-line memory module; access 64 bits)



SO-DIMM (small outline DIMM)



Note) SIMM (single in-line memory module; access 32 bits)

Image: Wikipedia

ROM

- **Read-only memory (ROM)**
  - A type of non-volatile memory
  - Useful for storing firmware (software that is rarely changed during its life)
    - e.g. Consumer electronics, Automobiles, BIOS (Basic Input/Output System)
- **Types**
  - **Mask ROM** (~ CD-ROM, printing)
    - Factory programmed ROM, but very low cost
  - **Programmable ROM (PROM)** (~ CD-R)
    - One-time programmable ROM (Alias: OTP-PROM)
    - A special equipment, ROM writers, rewrite PROM/EPROM/EEPROM.
  - **Erasable Programmable ROM (EPROM)**
    - Its contents are erased by strong ultraviolet (UV) light. (Alias: UV-EPROM)
  - **Electronically Erasable Programmable ROM (EEPROM)** (~ CD-RW)
    - It is composed of floating-gate transistors for rewriting.
    - Its modern alternative: Flash memory for high speed and high density

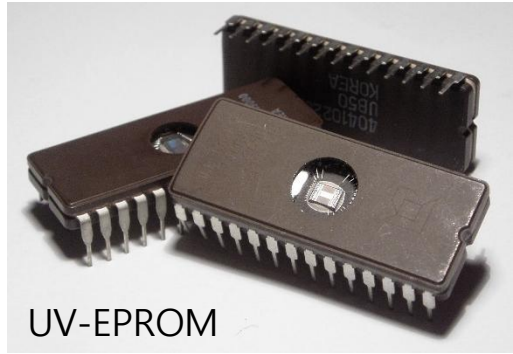

UV-EPROM

Image: Wikipedia

ROM

- **ROM using a decoder and OR gates**
  - 32 x 8-bit ROM array (total 32 bytes)

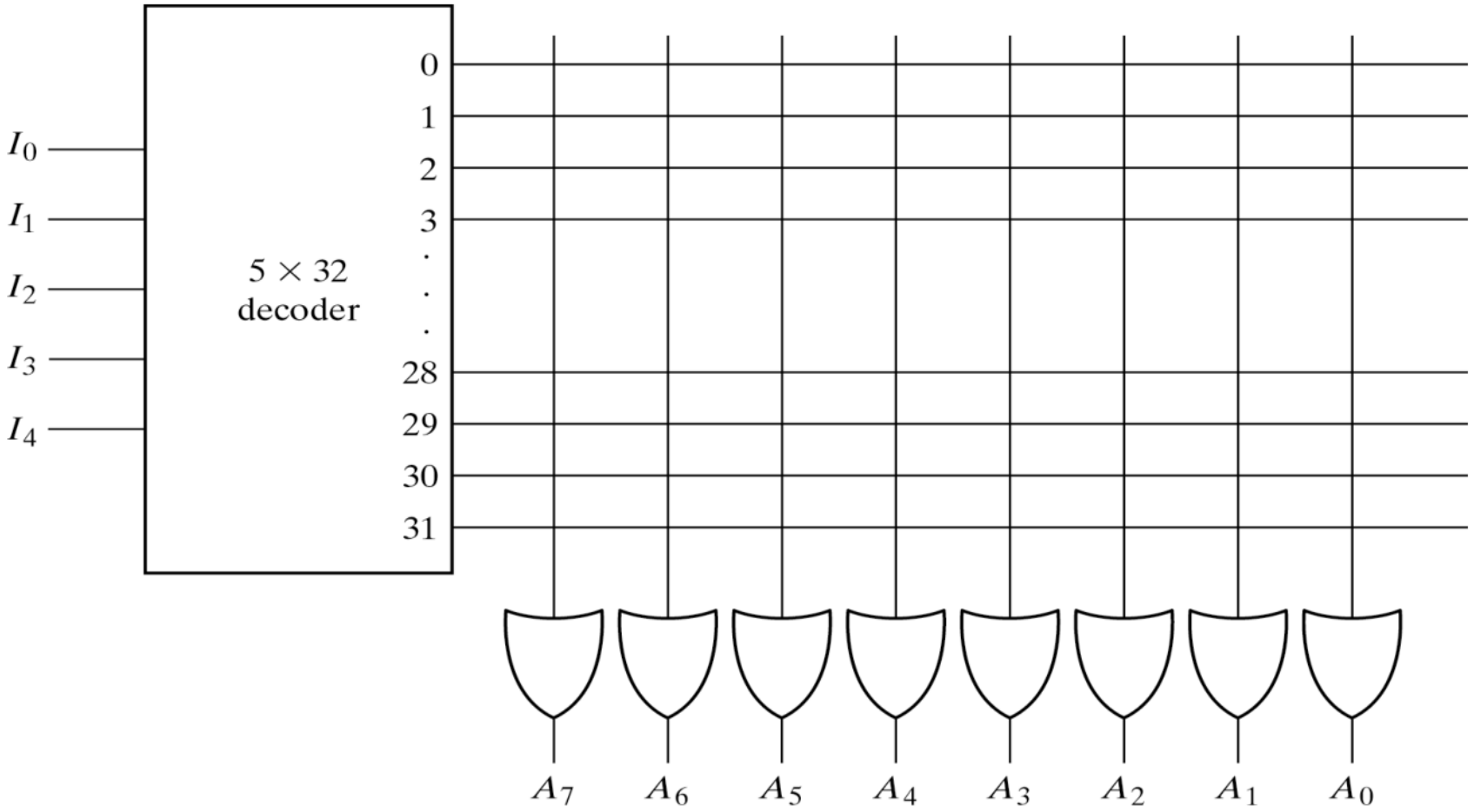


Image: Amirali Baniasadi, Lecture 13 slides of *Digital Design* (CENG 241), 2015

ROM

- **ROM using a decoder and OR gates**
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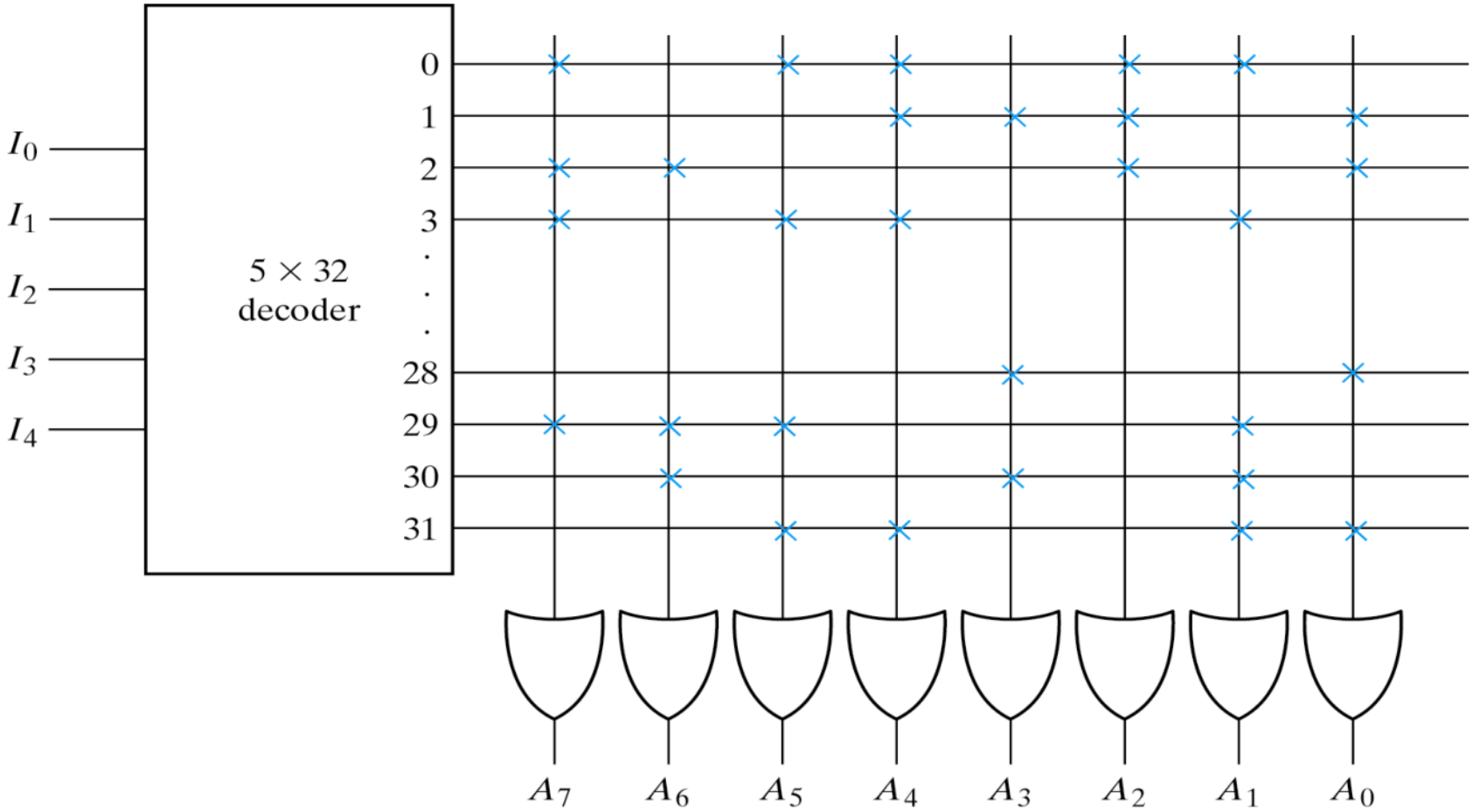
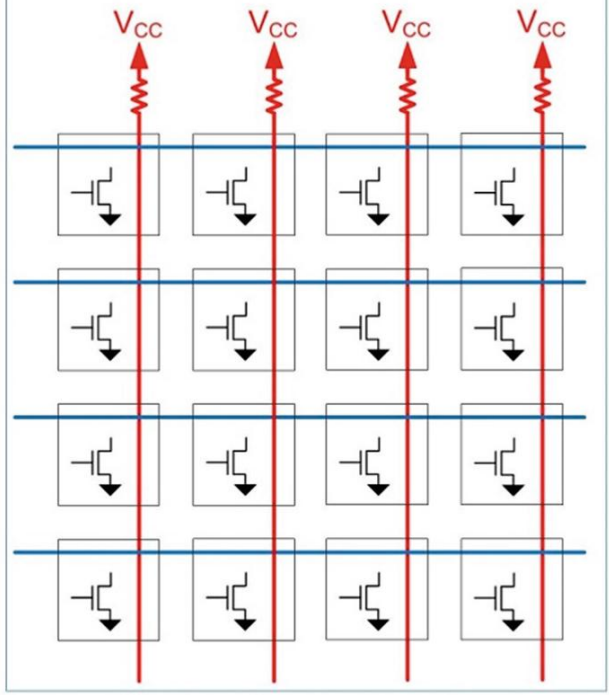


Image: Amirali Baniasadi, Lecture 13 slides of *Digital Design* (CENG 241), 2015

Mask ROM

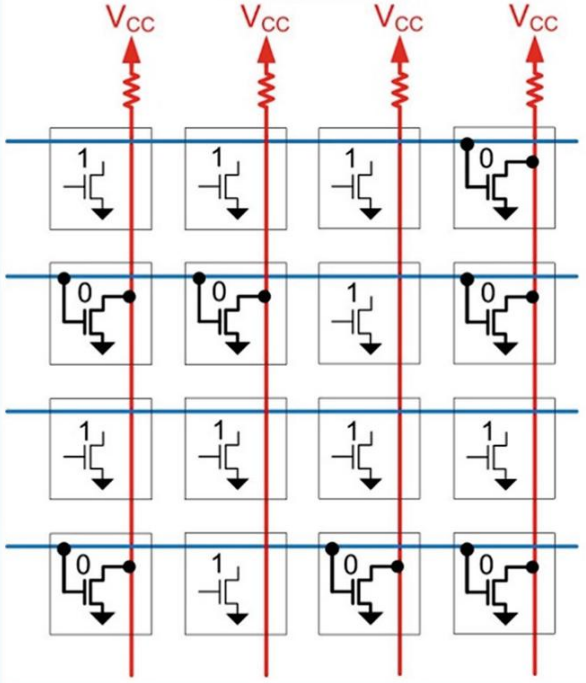
The MROM device is partially fabricated to contain all of the NMOS transistors, word lines, bit lines, pull-up networks and interfacing circuitry. The fabrication is then stopped prior to connecting the gate and drain terminals of the NMOS transistors to the word and bit lines. The device is considered “unprogrammed” at this point.



Unprogrammed

When the data to be stored is provided by the customer, the connections between the NMOS transistors and the word/bit lines are implemented in order to create the desired 0's.

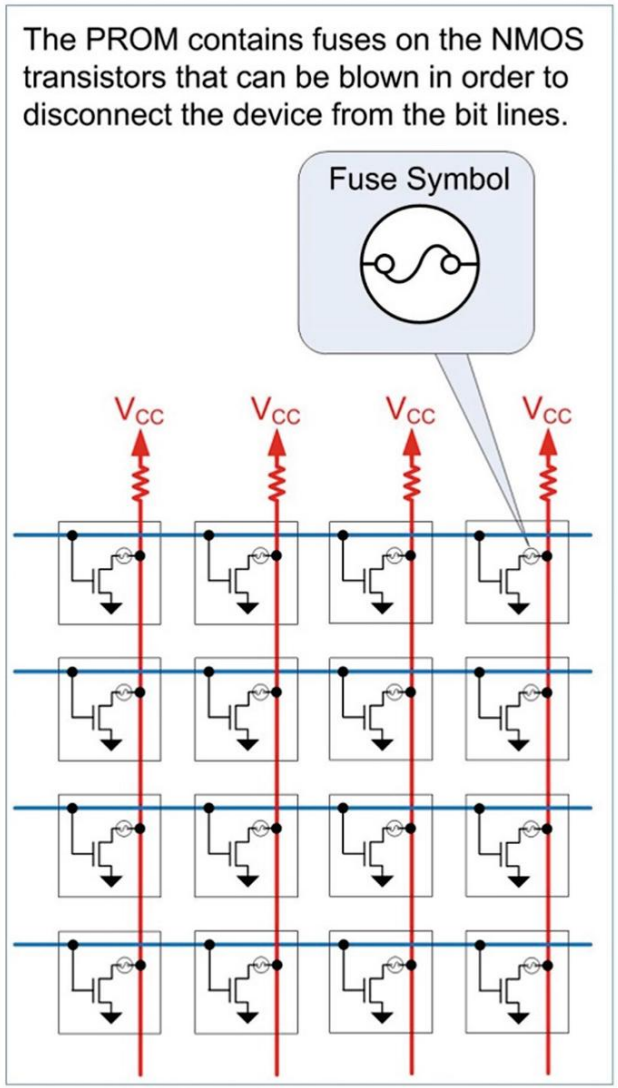
Address	Data
0	1 1 1 0
1	0 0 1 0
2	1 1 1 1
3	0 1 0 0



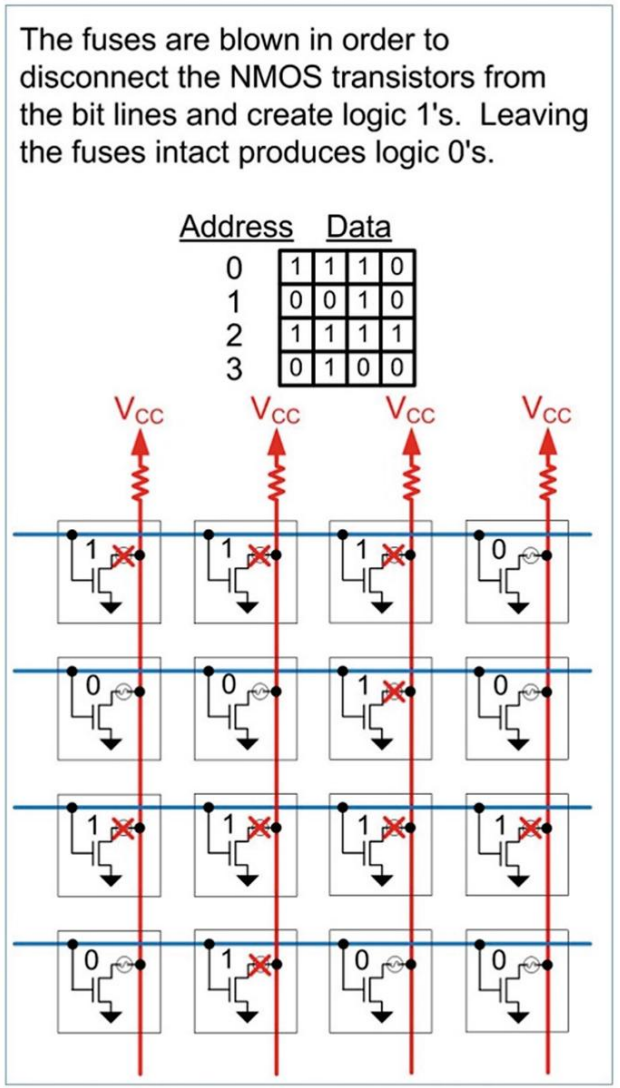
Programmed

Image: [Brock J. LaMeres, "Memory" (Book Chapter), Springer, 2019]

Programmable ROM



Unprogrammed

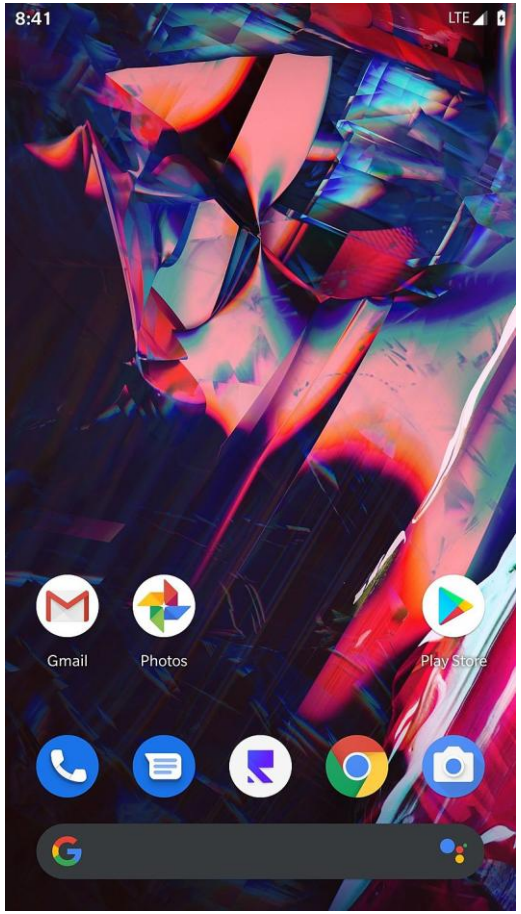


Programmed

Image: [Brock J. LaMeres, "Memory" (Book Chapter), Springer, 2019]

Note) Custom ROM?

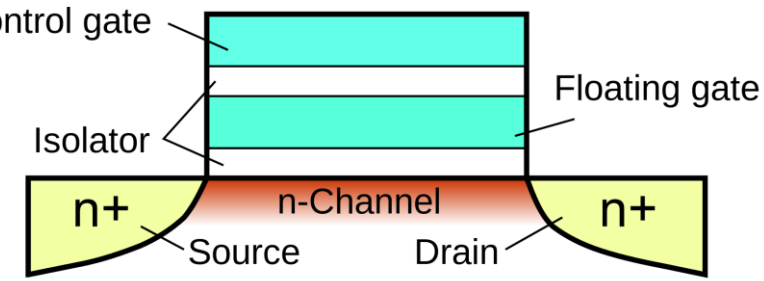
- Please use a ~~custom ROM~~ custom firmware as a more correct term.



Paranoid Android

Flash Memory

- **Flash memory**
  - A type of non-volatile memory
  - Electrically erasable and reprogrammable
    - It is composed of floating-gate transistors similarly to EEPROM.
  - Flash memory can **erase/write specific blocks** (or pages), but EEPROMs needs to erase/write the entire contents.



- **NAND flash memory** vs. **NOR flash memory**

Attribute	NAND	NOR
Main application	**Data storage** (USB drive, memory card, SSD)	**Code execution** (firmware, bootloader)
Storage capacity	**High (dense)**	Low
Cost per bit	Low	High
Active power	Low	High
Standby power	High	Low
Write speed	Fast	Slow
Read speed	Slow	**Fast**
Execute in place (XIP)	No	**Yes**
Reliability	Low	High

Image: Wikipedia

NOR Flash Memory

- **NOR flash memory**
  - It is composed of *individual* NOR cells.
  - It allows *random access* (~ RAM/ROM). → Execute in place (XIP).

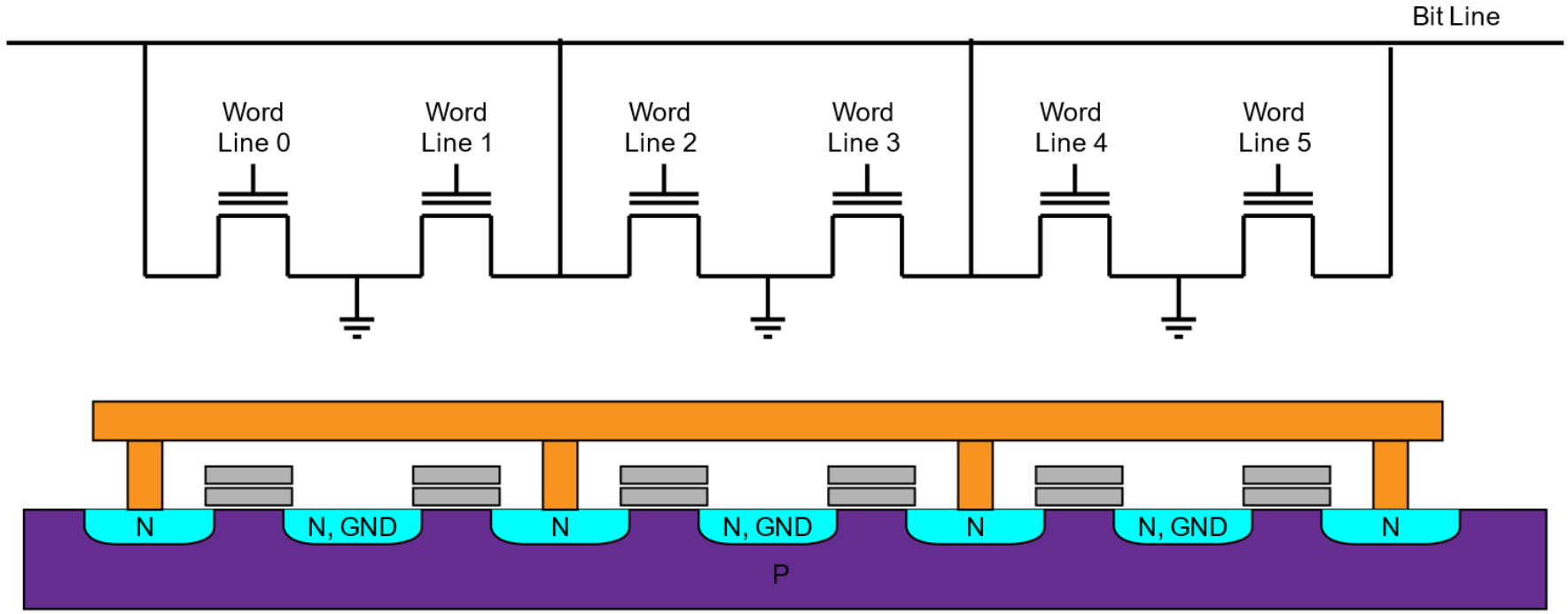


Image: Wikipedia

NAND Flash Memory

- **NAND flash memory**
  - It is composed of a *series* of NAND cells. → High storage capacity
  - It allows only *page access*. → Slow read speed
    - Note) HDD is good at *sequential access*.

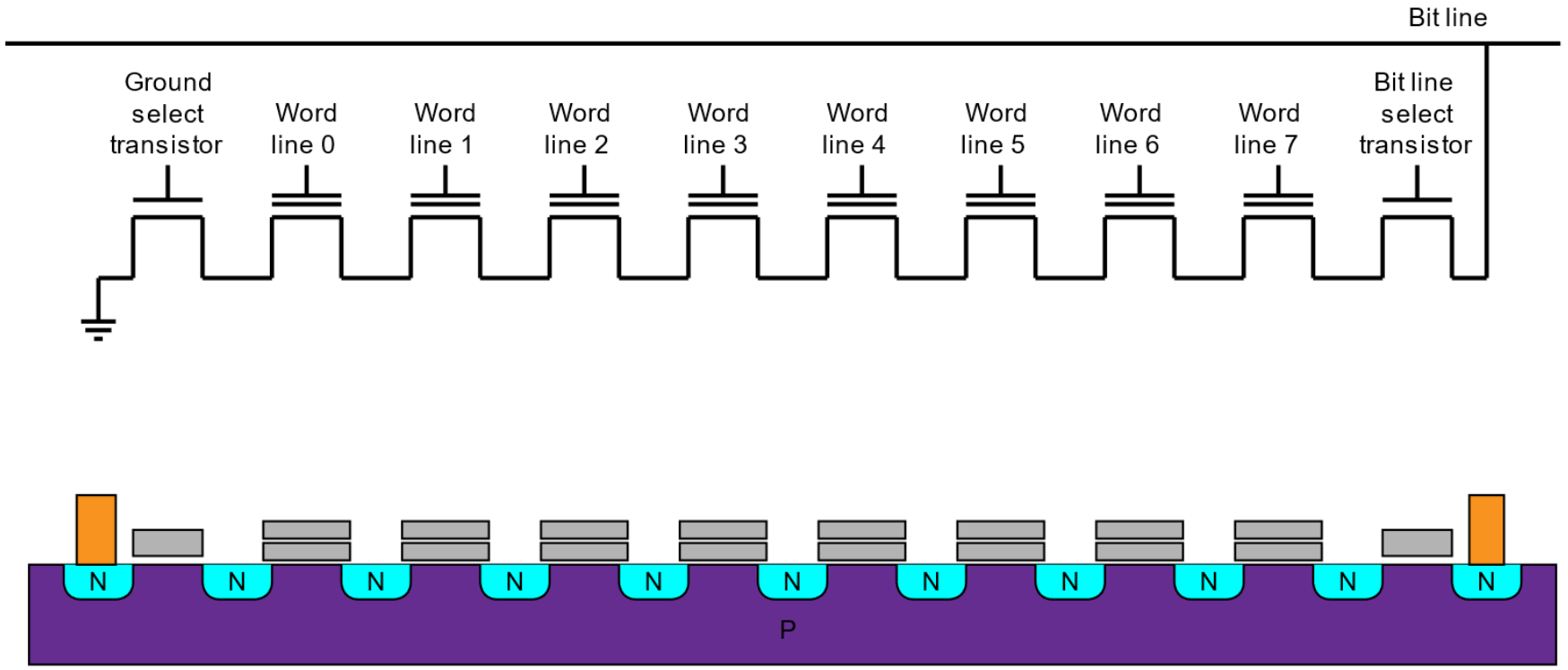


Image: Wikipedia

HDD and SSD

- **Hard disk drive (HDD)**
  - An *electro-mechanical* storage device using *magnetics* (magnetic storage)
  - It can store and retrieve data using a spinning magnetic disk.

Note) optical storage

- **Solid-state drive (SSD)**
  - A non-volatile storage device using *only electronic circuits* (solid-state storage)
  - It can store and retrieve data using NAND flash memory.

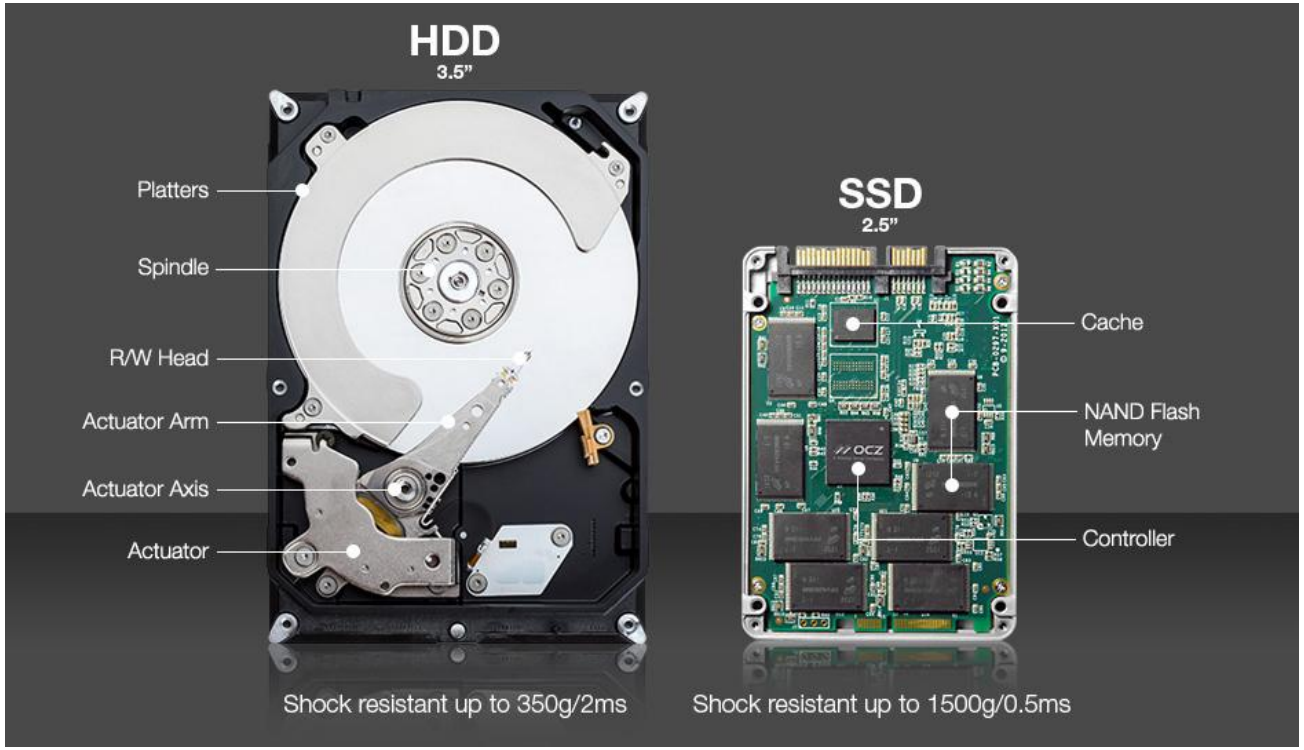


Image: Data Symposium

HDD and SSD

- **HDD** vs. **SSD** [more explanation]

(IC: Integrated Circuit; 집적회로 in Korean)

Attribute	HDD	SSD (IC, no moving parts)
Price per capacity	**Cheap** (3 cents/GB, 2018)	Expensive (30 cents/GB, 2018)
Storage capacity	**Large**	Small
Storage areal density	Less dense (1.2 Tb/in^2)	**Dense** (2.8 Tb/in^2)
Reliability – data retention	**Many years**	10 years
Reliability – longevity	9-11 years	**Many years**
Start-up time	Several seconds (start to move)	**A few milliseconds**
Sequential access	200 MB/s	**3500 MB/s**
Random access	2.9 – 12 ms	**Less than 0.1 ms**
Read/write time symmetry	**Slightly slower for writing**	Significant slower for writing
Temperature lower limit	High (above 0 °C)	**Low** (above -55 °C)
Susceptibility to impact	Weak	**Strong**
Susceptibility to magnetics	Weak	**Strong**
Size and weight	Large and heavy	**Small** (e.g. M.2) **and light**
Acoustic noise	High	**Low**
Power consumption	High	**Low**

Memory Hierarchy

- **Memory hierarchy** based on *response time*

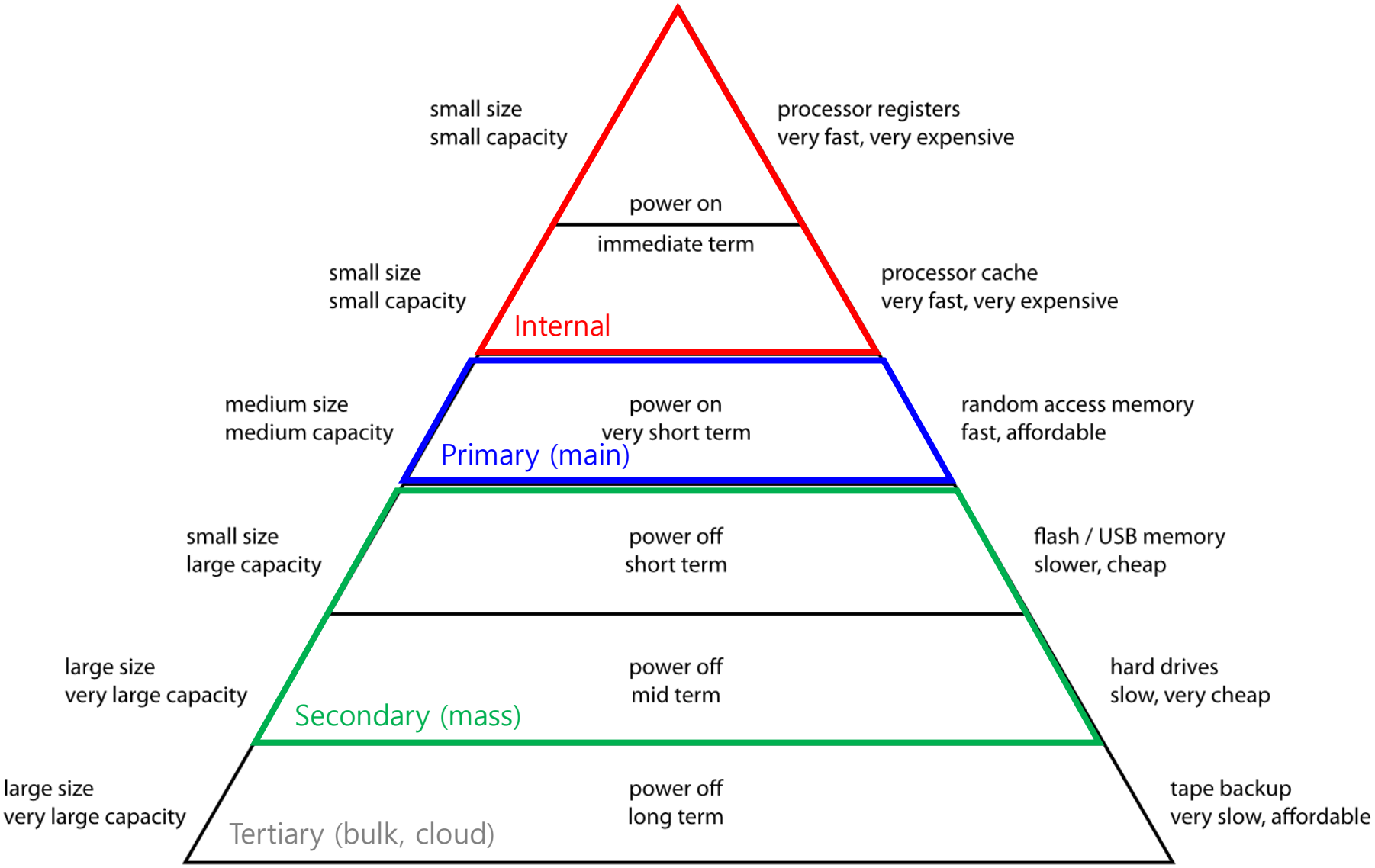


Image: Wikipedia

Summary) Memory/Storage Hierarchy (Note: Volatile)

- Internal memory (inside of CPU)
  - Registers and **SRAM** (for caches)
- Primary memory (directly accessed by CPU; alias: main memory)
  - **DRAM** (SDRAM and DDR SDRAM)
  - **ROM** (Mask ROM, PROM, EPROM, and EEPROM)
  - **NOR flash memory**
- Secondary storage devices
  - **NAND flash memory**
  - **SSD**, **HDD**
- Tertiary storage devices
  - e.g. Nearline storage, cloud storage